
Closing the Frequency Gap: Deploying Vision-Language-Action Models on High-Rate Humanoid Control

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Abstract

1 Foundation Vision-Language-Action (VLA) models unlock open-vocabulary ma-
2 nipulation, but measured inference rates remain incompatible with the 100 Hz
3 joint loops required by humanoid robots such as the Unitree G1 (29 DoF). We
4 study this *frequency gap* as a Physical AI deployment bottleneck: on a DGX
5 Station (Tesla V100), live UnifoLM-VLA-Base profiling on `G1_Clean_Table`
6 yields a $\sim 57\times$ gap to 100 Hz (PyTorch mean **570.7 ms / 1.752 Hz**; Nsight NVTX
7 **593.7 ms / 1.684 Hz**). We keep the VLA as a sparse language-conditioned in-
8 tent generator and insert a fixed Echo State Network (ESN) reservoir that bridges
9 held joint targets to smooth 100 Hz commands; only a linear readout is trained
10 by ridge regression. On `G1_Dex1_Wipe_Table` (200 episodes, ~ 0.70 h; train 0–
11 159 / held-out 160–199), a CUDA ESN ($\alpha=0.3$, $\lambda=1$) reaches held-out open-loop
12 RMSE $2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$ **rad** ($\sim 5.6\times$ better than linear upsampling) and
13 MuJoCo dataset-oracle RMSE $3.00 \times 10^{-3} \pm 3.56 \times 10^{-3}$ **rad** (grasp/task success
14 **100%**; mean table-contact **74.5%**). Live dual-process control clears 100 Hz on av-
15 erage (ESN mean 6.92 ms / 145 Hz) with GPU-contention tails above 10 ms. Live
16 wipe trials succeed under a disclosed geometric press prior for both ESN and ZOH;
17 under that tied success regime, ESN joint RMSE is $\sim 14\times$ lower than ZOH (0.514
18 vs. 7.17 rad), while open-loop ZOH remains $50\times$ jerkier than linear—motivating
19 S2R/DIRT transfer rather than claiming zero-shot press-from-vision.

20 1 Introduction

21 1.1 Motivation: Why VLAs Break on Humanoids

22 Foundation VLAs such as OpenVLA Kim et al. [2025], π_0 Black et al. [2024], and Unitree UnifoLM-
23 VLA Unitree Robotics [2026] map language and vision to robot actions and have unlocked open-
24 vocabulary manipulation on fixed-base arms. Humanoid whole-body control is a different regime Po-
25 ria et al. [2025]: the Unitree G1 exposes a **29-DoF** joint interface that must be served near **100 Hz**
26 (10 ms) for balance-preserving tracking, contact-rich wiping, and safe impedance behavior. Under
27 a naive zero-order hold (ZOH), each sparse VLA target is frozen for tens of control ticks; staircase
28 joint trajectories inject high jerk and fail contact tasks even when the semantic plan is correct. We
29 call this mismatch the *frequency gap*: the VLA may be “right” about *what* to do, yet too slow for
30 *how* the humanoid must move.

VLA + ESN hybrid control for Unitree G1 (29-DoF)

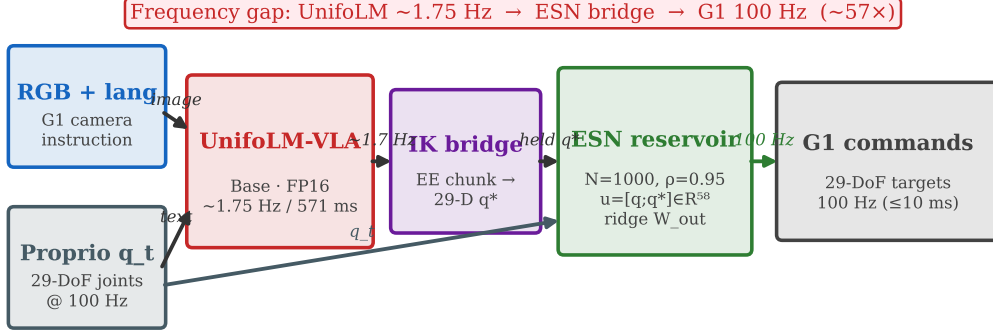


Figure 1: VLA+ESN architecture for Unitree G1 (29 DoF). UnifoLM-VLA-Base runs at ~ 1.75 Hz on a DGX V100; the ESN bridge closes the $\sim 57\times$ gap to the 100 Hz joint loop.

This paper contributes a systems perspective to Physical AI deployment rather than a new zero-shot foundation policy. The control layer must be *cheap* and preferably trained without backpropagation through time, because lab/edge GPUs are already saturated by vision–language backbones. Classical upsamplers (ZOH, linear interpolation, joint PID) ignore proprioceptive state between VLA updates and cannot learn demonstration-consistent dynamics.

1.2 Approach and Scope

We insert a leaky Echo State Network (ESN) between measured VLA intents (~ 1.7 Hz on UnifoLM-VLA-Base) and joint commands (100 Hz). The VLA supplies sparse targets q_{VLA}^* ; the reservoir supplies reflexive temporal filtering with a ridge-trained linear readout and *fixed* recurrent weights. Training and evaluation use Unitree’s public G1_Dex1_Wipe_Table demonstrations (Sec. 4.1).

Contributions. (i) Dual-instrument UnifoLM-VLA-Base latency quantifying a $\sim 57\text{--}59\times$ gap to 100 Hz; (ii) CUDA ESN with $u(t) = [q_t; q_{\text{VLA}}^*] \in \mathbb{R}^{58}$, ridge readout, declared train/held-out split, and a per-task checkpoint library for the UnifoLM G1 suite; (iii) GIL-free dual-process MuJoCo+bridge timing under live UnifoLM (ESN vs. ZOH / linear / PID); (iv) MuJoCo wipe-table *dataset-oracle* evaluation on all 40 held-out episodes; (v) Live UnifoLM wipe-success trials with interactive cloth, including disclosed geometric priors and failure modes; (vi) S2R ZMQ deploy package (29-DoF defaults) for DIRT-guided transfer (in progress).

2 Related Work

VLA surveys highlight bimanual and whole-body settings as hard cases Anonymous [2026]. Flow matching accelerates action generation Black et al. [2024]; recurrent bridges such as xLSTM help long-horizon memory Haresh et al. [2026]. Diffusion action chunking trades compute for smoothness; reservoir computing offers fast, training-light dynamics Jaeger [2001], Cao et al. [2023]. We combine a frozen reservoir with a language-conditioned VLA rather than replacing either.

3 Method

3.1 System Architecture

Figure 1 summarizes the hybrid stack: UnifoLM emits sparse EE intents, an IK bridge lifts them to 29-D joint targets, and a fixed ESN reservoir with ridge readout produces 100 Hz commands conditioned on proprioception.

- **Perception / VLA:** UnifoLM-VLA-Base emits **23-D end-effector** action chunks; a Jacobian **IK bridge** maps EE poses to 29-D joint targets.

Table 1: Measured results on DGX Station (V100), 5 Aug 2026 campaign (held-out wipe unless noted).

Stage	Metric	Value / notes
PyTorch	Mean latency / rate / gap	570.7 ms / 1.752 Hz / $57.1 \times (n=100)$
Nsight	NVTX mean / gap	593.7 ms / 59.4 $\times$; GEMM + H2D heavy
ESN ridge	Held-out RMSE	$2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$ rad ($\alpha=0.3$, $\lambda=1$)
Offline	ESN vs. linear	$\sim 5.6 \times$ lower RMSE; ZOH jerk $50 \times$ linear
Dual-process	Live ESN mean / rate	6.92 ms / 145 Hz (steady max 12.6 ms)
MuJoCo oracle	RMSE / grasp / contact	3.00×10^{-3} rad; 100% / 74.5% ($n=40$)
Live wipe	Contact / task (ESN & ZOH)	100% / yes (same <code>press_table</code> prior; 30 s)
Live wipe	Joint RMSE (ESN vs. ZOH)	0.514 vs. 7.17 rad ($\sim 14 \times$ lower)

Table 2: Open-loop held-out upsample quality on wipe eps 160–199 ($n=40$). Jerk ratio is relative to linear (ref. = 1).

Method	Joint RMSE (rad)	Jerk ratio
ZOH	$5.39 \times 10^{-2} \pm 9.44 \times 10^{-3}$	$50 \times$
Linear	$1.55 \times 10^{-2} \pm 3.93 \times 10^{-3}$	1 (ref.)
PID	$7.14 \times 10^{-2} \pm 1.19 \times 10^{-2}$	$1.43 \times$
ESN (ours)	$2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$	$1.44 \times$

- 61 • **ESN bridge:** Leaky reservoir

$$x(t) = (1 - \alpha)x(t-1) + \alpha \tanh(W_{\text{res}}x(t-1) + W_{\text{in}}u(t)),$$

- 62 with sparse W_{res} ($N=1000$, $\rho=0.95$, sparsity 0.9) and readout $y(t) = W_{\text{out}}[x(t); u(t)]$ fit by
63 ridge regression on 100 Hz joints with VLA tokens held for 50 ticks ($u \in \mathbb{R}^{58}$, $y \in \mathbb{R}^{29}$). We
64 sweep $\alpha \in \{0.05, 0.1, 0.3\}$ and $\lambda \in \{10^{-4}, 10^{-2}, 1\}$ and select by RMSE $\times$ jerk score.

- 65 • **S2R / DIRT layer:** Parameterizes sim–real discrepancy for controller and domain ablations (ESN
66 vs. passthrough ZOH/raw).

67 3.2 Baselines

- 68 (1) Pure VLA ZOH; (2) VLA + linear interpolation; (3) VLA + PID; (4) VLA + ESN (ours).

69 4 Experiments

70 4.1 Dataset, Setup, and Metrics

71 All ESN training, offline upsample baselines, and MuJoCo wipe-oracle evaluations use the Hug-
72 ging Face dataset G1_Dex1_Wipe_Table (unitreerobotics; train split): **200** episodes (75,909
73 frames; ~ 0.70 h), train episodes 0–159 / held-out 160–199. Detailed wipe claims are restricted to
74 this corpus; the stack supports a per-task ESN library over the UnifoLM G1 suites. Robot / hardware:
75 Unitree G1 (29 DoF); NVIDIA DGX Station, Tesla V100-DGXS-32GB. Profiling: live UnifoLM-
76 VLA-Base on G1 Clean-Table (FP16). Metrics: task success (grasp / wipe contact), joint RMSE,
77 trajectory jerk, control Hz / step latency, GPU memory, parse failure rate.

78 4.2 Measured Status

79 Table 1 summarizes the campaign; Table 2 breaks out open-loop held-out RMSE and jerk (the
80 differentiator when live task success ties); Table 3 shows the same ESN bridge on two additional
81 UnifoLM G1 skills (open-loop only). Figure 2a shows the measured frequency gap; Fig. 2b open-
82 loop held-out RMSE; Fig. 3a live dual-process timing; Fig. 3b oracle held-out wipe metrics; Fig. 4
83 the contact ladder.

Table 3: Per-task ESN open-loop held-out joint RMSE (same $\alpha=0.3$, $\lambda=1$, 80/20 episode split). Not live multi-task success.

Task	Held-out n	Joint RMSE (rad)
wipe_table (primary)	40	$2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$
clean_table	40	$3.68 \times 10^{-3} \pm 6.72 \times 10^{-3}$
stack_block	40	$4.25 \times 10^{-3} \pm 2.84 \times 10^{-3}$

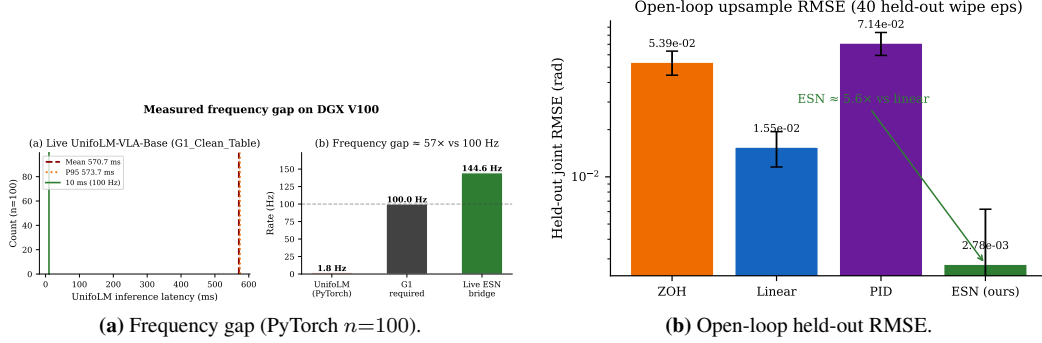


Figure 2: Latency gap and open-loop upsample baselines on held-out wipe episodes.

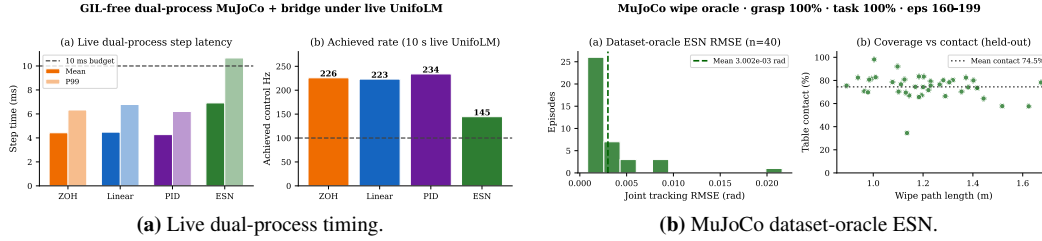
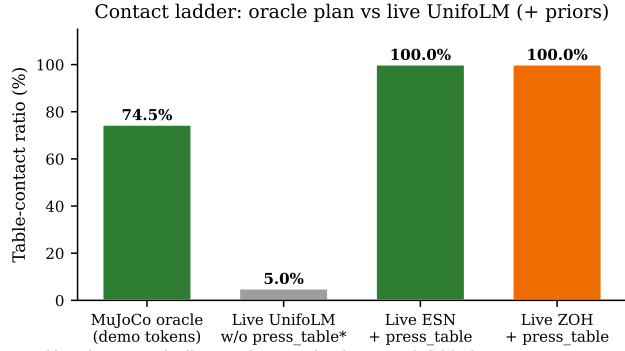


Figure 3: Live dual-process rates and held-out MuJoCo oracle wipe metrics.



* ~5% without press_table is the measured collapse mode reported in the paper (cloth lift); live rows use proximity-synthetic Dex1 + press_table.

Figure 4: Contact ladder: oracle demo tokens (74.5%) vs. live UnifoLM without `press_table` (~5% collapse) vs. live ESN/ZOH with disclosed geometric press prior (100%).

84 5 Discussion and Limitations

85 **Frequency gap.** Two instruments cross-validate $\sim 57\text{--}59\times$ to 100 Hz; both attribute GPU time to
86 large GEMMs, with Nsight additionally highlighting H2D pressure.

87 **Data scope.** One wipe task for closed-loop claims, ~ 0.7 h of Dex1 demos, episode-level 80/20
88 split. Open-loop held-out joint RMSE stays in the 10^{-3} rad regime on `clean_table` and
89 `stack_block` as well (Table 3)—evidence that the bridge is not wipe-only—but this is *not* live
90 multi-task or zero-shot competence.

91 **ESN, timing, and honest live findings.** Held-out open-loop ESN RMSE is $\sim 5.6\times$ better than
92 linear, with jerk near the linear reference ($1.44\times$) while ZOH is $50\times$ jerkier (Table 2). Live dual-
93 process ESN clears 100 Hz mean (145 Hz) with GPU-contention tails above 10 ms. Oracle wipe
94 succeeds when demo tokens drive the ESN. Matching live UnifoLM wipe trials attach under a
95 proximity-synthetic grasp; without a table-plane press prior the VLA lifts the cloth and contact
96 collapses ($\sim 5\%$). Enabling `press_table` restores 100% contact and task success for *both* ESN
97 and ZOH—a geometric wipe prior, not learned press-from-vision. Under that tied success regime,
98 ESN still tracks joints $\sim 14\times$ tighter than ZOH (0.514 vs. 7.17 rad RMSE over 30 s), which is the
99 systems reason to prefer the reservoir bridge even when binary task gates do not separate methods.
100 We treat the press-prior dependence as a deployment / negative-result style observation aligned with
101 the workshop’s interest in real-world failure analysis.

102 6 Conclusion

103 We presented a VLA+ESN architecture targeting the $\sim 57\times$ gap between measured UnifoLM-VLA-
104 Base inference and G1 100 Hz control. The ESN closes rate, tracking, and jerk gaps under declared
105 data splits; when live wipe success is tied by a disclosed geometric prior, joint RMSE still favors
106 ESN over ZOH by $\sim 14\times$. S2R/DIRT transfer on Jetson→G1 is the next step toward Physical AI
107 deployment claims beyond simulation.

108 References

- 109 Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov,
110 Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake,
111 Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-
112 action model. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, pages 2679–2713, 2025.
- 113 Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy
114 Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot
115 control. *arXiv preprint arXiv:2410.24164*, 2024.
- 116 Unitree Robotics. UnifoLM-VLA-0: Vision-language-action for humanoid manipulation. <https://github.com/unitreerobotics/unifolm-vla>, 2026. Models and datasets on Hugging Face (unitreerobotics).
- 117 Soujanya Poria, Navonil Majumder, Chia-Yu Hung, et al. 10 open challenges steering the future of vision-
118 language-action models. *arXiv preprint arXiv:2511.05936*, 2025.
- 120 Anonymous. Vision-language-action (vla) models for unmanned aerial robotics and bimanual manipulation: A
121 review. *Preprints.org*, 2026. doi:10.20944/preprints202604.0664.v1.
- 122 Sanjay Haresh, Daniel Dijkman, Apratim Bhattacharyya, and Roland Memisevic. Notes-to-self: Scratchpad
123 augmented vlas for memory dependent manipulation tasks. *arXiv preprint arXiv:2602.21013*, 2026.
- 124 Herbert Jaeger. The “echo state” approach to analysing and training recurrent neural networks. *GMD Report*,
125 (148), 2001.
- 126 Ruizhi Cao, Chun Guan, Zhongxue Gan, and Siyang Leng. Reviving the dynamics of attacked reservoir com-
127 puters. *Entropy*, 25(3):515, 2023.